

* Expectation Decider *

Q-1) what is probability?

Ans

- Probability is mathematical measure of how likely it is that an event will happen.

Q-2) key terminology of probability.

Ans Experiment: any action like tossing a coin or rolling a die that can be repeated and has a defined result.

- outcome: A single result or observation from an experiment
- Sample space: The complete set of all possible outcomes for an experiment, usually written as S
- Event: Any specific outcome or combination of outcomes from the sample space.
- Probability: A numerical value between 0 and 1 that tells us how likely an event is to occur.

Q-3) Three Example of event from dataset

Ans 1) Student passes the final exam.

2) student studied more than 15 hours per week.

3) student has at least 75% attendance AND passed the final Exam.

Q-4) Types of events

Any • Empirical probability = student passes.

- Total students = 200
- students who passed = 69

$$P(A) = \frac{\text{number of students pass}}{\text{total students}} = \frac{69}{200} = 34.5\%$$

• Theoretical probability - Group discussion response

$S = \{\text{Yes, No}\}$

$B = \{\text{Yes}\}$

$$P(B) = \frac{1}{2}$$

= 50%

Q-5) define a random variable for the event "Number of students passing the final exam out of 3 randomly selected student."

Any $X =$ number of students who passes the final exam among the 3 selected students.

$$X \in \{0, 1, 2, 3\}$$

Q-6) construct its probability distribution table.

Any $X =$ number of students who passes among 3 selected students

$$\rightarrow P(\text{Pass}) = 69/200 = 0.345$$

$$P(\text{fail}) = 1 - 0.345 = 0.655$$

- X follows a binomial Distribution:

$$n=3$$

$$p=0.345$$

The binomial PMF: $P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$

So,	X	0	1	2	3
	$P(X)$	0.2810	0.440	0.2339	0.0411

Q-7) Find the mean and variance of this random variable.
Ans X = number of students passing among 3 selected students.

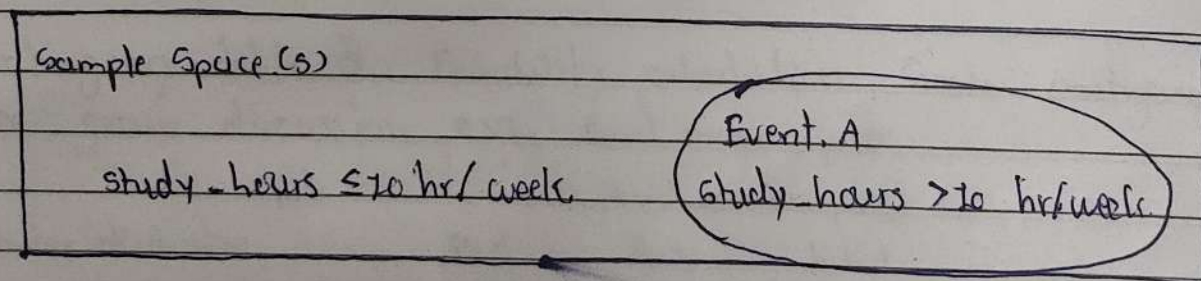
- $n=3$, $p=0.345$, $q=1-p=0.655$

• Mean $EX = np = 3 \times 0.345 = 1.035$

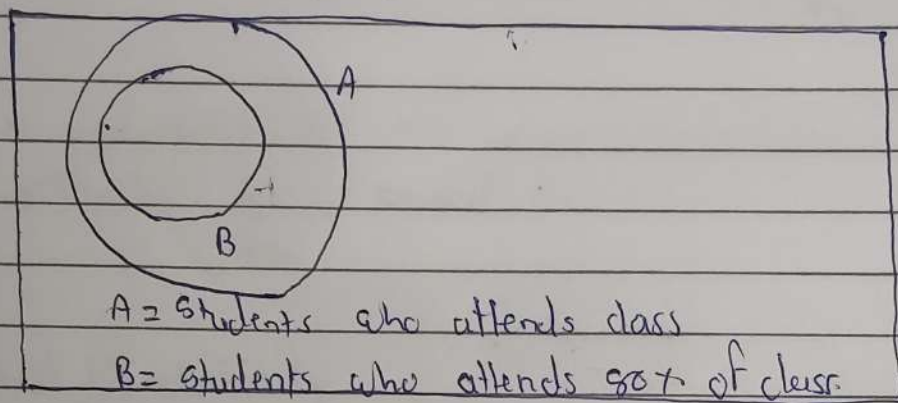
• Variance $EX^2 = npq = 3 \times 0.345 \times 0.655 = \cancel{0.26} 0.677025$

Q-8) Venn diagram in probability. draw a Venn diagram showing

- students who study more than 10 hours/week.

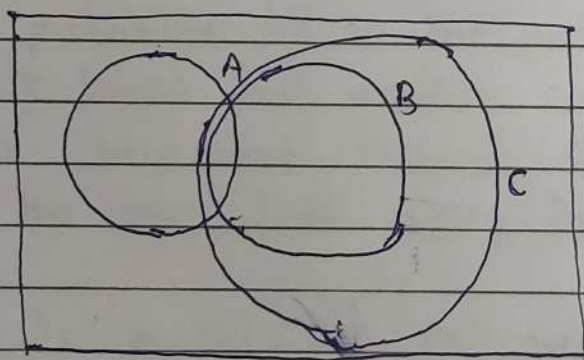


- students who attend more than 80% of classes.



- Overlap showing students who satisfy both condition.

sample spaces)



A = students who study > 10hr
B = students study attend 80% class
C = student attend class

Q-4) Contingency table & Probability calculations, Create contingency table for group discussion vs. Final exam pass.

Any

Group discussion	Final exam pass		
	Fail	Pass	total
No	70	19	89
yes	61	50	111
total	131	69	200

1. Joint probability

$$p(\text{Group Discussion} = \text{Yes AND Pass}) = \frac{50}{200} = 25\%$$

2. marginal probability

$$p(\text{Pass}) = \frac{69}{200} = 34.5\%$$

3. Conditional probability

$$p(\text{Pass} | \text{yes}) = \frac{50}{111} = 45.05\%$$

Q-10) Cond Interpret conditional probability result in plain language.
Any conditional probability tells us the probability of an event when we already know that another condition is true. The word "given" is important because it reduces the sample space to only the cases that satisfy that condition.

Ex

- 111 students participated in group discussion.

- among these 111 students, 50 passed the final Exam.

$$50 \quad p(\text{Pass} | \text{yes}) = \frac{50}{111} = 45.05\%$$

So, instead of considering 200 students, we consider only 111 students who participated in group discussion.

Q17) identify if "participating in group discussion" and "passing exam" are independent, dependent or mutually exclusive events.

Justify your answer

Ans) A = students participated in group discussion

B = student passes the final exam

$$P(A) = \frac{111}{200} = 0.555$$

$$P(B) = \frac{69}{200} = 0.345$$

$$P(A|B) = \frac{50}{200} = 0.25$$

$$\begin{aligned} \text{if A and B are independent } P(A|B) &= P(A)P(B) \\ &= (0.555)(0.345) \\ &= 0.191 \\ P(A|B) &\neq P(A)P(B) \end{aligned}$$

So A and B are dependent

- Two events are mutually exclusive if they cannot occur together meaning $P(A|B) = 0$
But $P(A|B) \neq 0$ so, events are mutually exclusive

Q18) Use the Bayes theorem to solve below problems.

Ans) 70% of students who pass had high attendance.

- AP = student pass

BF = student fails

H = student has high attendance (>80%)

$$P(H|P) = 0.70$$

$$P(H|F) = 0.40$$

$$P(H) = 0.60$$

$$P(H) = P(H/P) \cdot P(P) + P(H/F) \cdot P(F)$$

$$= \cancel{P(F)}$$

$$0.60 = 0.70 P(P) + 0.40 (1 - P(P))$$

$$0.60 = 0.70 P(P) + 0.40 - 0.40 P(P)$$

$$0.20 = 0.30 P(P)$$

$$P(P) = 66.67\%$$

$$P(F) = 33.33\%$$

Now,

$$- P(H \cap P) = P(H/P) P(P) = 0.70 \times 0.66 = 0.466$$

$$- P(H \cap F) = 0.40 \times 0.333 = 0.133$$

$$\text{So } P(H) = P(H \cap F) + P(H \cap P)$$

$$= 46.67 + 13.33$$

$$= 60\%$$